Change the Plan

The plan is where you apply transactions and all the data enrichment. You have two types of plans:

- **Deployable plans**: can be used in production.
- **Non-deployable plans**: allow for more flexibility on data exploration, but are not deployable in production.

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If you need to change the out-of-the-box metrics / fields, create a copy of the original plan first, and make your changes there.

There are five blocks in the out-of-the-box Cards plan, and they are processed sequentially.

- **List Enrichment**: picks up values in the list, if the entity exists in that list.
    - For example, if `card_id` exists in PosNeg Listed Cards and has a `decline` value, then `posneg_card_value` is `decline`.
- **Derived Fields / Metrics Based Fields**: enriches the data using fields/metrics created previously.
    - This is where you can use user-defined functions (UDFs), implement conditional logic with if/else statements, update fields, and delete fields as needed.
- **Metrics**: creates metrics based on a window.

Create a new metricâ

In the side panel, go to **Data Science**.

1. In the **Plans** widget, open the plan where you want to create a new metric.

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   Never change out-of-the-box metrics.

2. Select the **Pencil** icon from the **Metrics** operator.
3. Select **Add new Metric** and fill in the information as explained in the following sections.

Group Byâ

- **Field(s)**: this is where you select the fields by which you want to group events for your metric.

For example, if you want each card to have a different value, you can group by `card_id` .

Windowâ

- **Expression**: determines how long the window is (e.g. 1 minute, 1 hour, 1 day, 1 week). Note that a metric with a window of less than or equal to 24 hours can be set as real-time metric (i.e. "Update on every event" in the **Edit Update Period** section). A metric with a window greater than 24 hours must be set as scheduled metrics (i.e. "Start of day").

- **Delay**: allows you to delay the metric for a certain time to exclude recent transactions to be included in the metric.

- **Is tumbling**: if enabled, refreshes metrics when they hit the window duration.

- **Filter**: ensures that only Authorization or pre-decided Info will enter this metric, as mentioned in [Understand the Workflow](). If you need additional filters, feel free to add them here.

    - **Cards**

```
( event_type == 'authorization' or (event_type == 'info' and info_type ==
'pre_decided') ) and (card_id ?? '' != '')
```

If you group by any entity other than `card_id` , replace `(card_id ?? '' != '')` with `(YOUR_ENTITY ?? '' != '')` .

    - **Transfers**

```
(event_type == 'transfer_initiation' or (event_type == 'info' and info_type == 'pre_decid
ed'))
```

Edit Update Periodâ

- **Update Period**: if it's a metric with a window of less than or equal to 24 hours, select **Update on every event**, otherwise select **Start of day**.

Aggregationâ

- **Aggregation type**: select how you would like to aggregate the metric.

For example, if you want to use `amount`, use `cardholder_bill_amount` for Cards.

Create snapshotsâ

After creating new metrics, you need to create a new snapshot to [deploy this plan]() to production.

Tick the Schema's **Export** box and name it.

This schema will contain the newly-created metrics, and will be used in your rules project.

New metric - practical exampleâ

Consider the scenario where you need to create a rule that declines transactions from any card that has spent over $1,000.00 on casino transactions over the last 3 hours. You can also assume casino transactions are identified as MCC = 7800. Note that, if you need to use a field that is not in Feedzai's schema, e.g. you have a boolean flag that already identifies gambling transactions in your system, then you need to contact Feedzai to add this custom field into Feedzai's schema.

This scenario involves creating a metric that will aggregate the amount for gambling transactions. This means you'll have to create a metric in "Cards Plan v4", the plan used in the out-of-the-box workflow:

1. In the side panel, go to **Data Science** and then, in the **Plans** widget, open "Cards Plan v4".
2. Using the **Pencil** open the **Metrics** operator.
3. Select **Add new Metric**.

Group Byâ

- **Field(s)**: `card_id`

Windowâ

1. **Expression**: 3 hours
2. **Filter**: (event_type == 'authorization' or (event_type == 'info' and info_type == 'pre_decided')) and (card_id ?? '' != '') and (mcc ?? '' == '7800')

What does ?? mean?

- ?? means coalesce. If card_id is null, it will fill with empty string.
- card_id ?? '' != '' simply means it must not be null or empty string.

Edit Update Periodâ

1. Since the window is less than or equal to 24 hours, it will be a real-time metric.

Aggregationâ

1. **Function**: sum
2. **Field**: cardholder_bill_amount

Name & Descriptionâ

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If you don't fill in the name, Pulse will automatically generate a metric name for you. Note that metric names should clearly indicate the grouping entities, aggregation, window, and possibly filter. It is also recommended that you define a prefix to indicate they are custom metrics (e.g. `c_`). Also, as a rule of thumb, name your metrics in lower case, and separate each word using underscores.

1. **Name**: `c_cardid_sum_amt_3h_mcc7800`

In this scenario, the name generated by Pulse is `cardid_sum_cardholderbillamount_3h`, but you can easily edit it directly in the **Name** field.

Go to the next building block, Change Rules Projects, to learn how your rules project picks up these metrics.